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Question **1**

Not answered

Marked out of 6.00

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible!

Select one or more:

- a. ICA requires at least as many observations as there are latent sources.
- b. The mixing matrix $\mathbf{U}$ must be orthogonal.
- c. ICA is scale-invariant and its components are not ranked.
- d. FastICA is an iterative algorithm that uses whitening and rotation.
- e. INFOMAX is an algorithm that measures independence by maximizing the entropy $\mathbf{H}(\mathbf{g}(\mathbf{y}))$, where $\mathbf{g}(\mathbf{y})$ is some elementwise function of the latent sources.
- f. Measures of independence are the mutual information between observations or the non-Gaussianity of the components.
- g. ICA is an unsupervised method of finding directions in feature space that are most statistically independent from each other.
- h. The latent variables $\mathbf{y}$ in ICA are assumed to be Gaussian and independent, such that $p(\mathbf{y}) = \prod_i^l p(y_i)$

Die Antwort ist falsch.

The correct answers are: ICA is an unsupervised method of finding directions in feature space that are most statistically independent from each other., ICA requires at least as many observations as there are latent sources., FastICA is an iterative algorithm that uses whitening and rotation., INFOMAX is an algorithm that measures independence by maximizing the entropy $\mathbf{H}(\mathbf{g}(\mathbf{y}))$, where $\mathbf{g}(\mathbf{y})$ is some elementwise function of the latent sources.
, Measures of independence are the mutual information between observations or the non-Gaussianity of the components., ICA is scale-invariant and its components are not ranked.

Question **2**

Not answered

Marked out of 6.00

Which statements about Kernel PCA are correct?

Multiple answers are possible!

Select one or more:

- a. A kernel is the outer product of feature maps in Hilbert space, $k(\mathbf{x}, \mathbf{y}) = \mathbf{\Phi}(\mathbf{x})\mathbf{\Phi}^T(\mathbf{y})$
- b. The mapping $\mathbf{\Phi}(.)$ is a linear mapping to a higher dimensional space which maximizes the variance in the data.
- c. Performing PCA in the feature space is equivalent to solving the eigenvalue problem for the Gram matrix.
- d. KernelPCA maps data to a higher dimensional space using kernel methods and performs PCA on the mapped data.
- e. KernelPCA requires the computation of the feature map $\mathbf{\Phi}(.)$
- f. The Gram matrix is given by $\sum_i^n \mathbf{\Phi}(x_i)\mathbf{\Phi}^T(x_i)$.
- g. KernelPCA is an extension of PCA to nonlinear projections.
- h. The Gram matrix is a square matrix which contains the kernel applied to all pairs of data points.

Die Antwort ist falsch.

The correct answers are: KernelPCA is an extension of PCA to nonlinear projections., Performing PCA in the feature space is equivalent to solving the eigenvalue problem for the Gram matrix.,
The Gram matrix is a square matrix which contains the kernel applied to all pairs of data points., KernelPCA maps data to a higher dimensional space using kernel methods and performs PCA on the mapped data.

Question **3**

Not answered

Marked out of 6.00

Select correct statements. Multiple choices might be possible.

- a. E-step tightens the lower bound of the log-likelihood
- b. M-step tightens the lower bound of the log-likelihood
- c. E-step maximizes the lower bound and increases the log-likelihood
- d. M-step maximizes the lower bound and increases the log-likelihood
- e. E-step: $w_{k+1} = argmax_w F(Q, w)$
- f. M-step: $w_{k+1} = argmax_w F(Q, w)$
- g. E-step: $Q_{k+1} = argmax_Q F(Q, w)$
- h. M-step: $Q_{k+1} = argmax_Q F(Q, w)$

Your answer is incorrect.

The correct answers are:
E-step tightens the lower bound of the log-likelihood,

M-step maximizes the lower bound and increases the log-likelihood,

M-step: \(w_{k+1} = argmax_{w} F(Q,w) \)

,

E-step: \(Q_{k+1} = argmax_{Q}F(Q,w) \)

Question 4

Not answered

Marked out of 7.00

Given two vectors:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

Compute the outer product matrix $\mathbf{A} = \mathbf{x}\mathbf{y}^T$.

Enter the (integer) values of the third row of $\mathbf{A}$ below:

- $A_{31} =$
  ✖
- $A_{32} =$
  ✖
- $A_{33} =$
  ✖

Question **5**

Not answered

Marked out of 7.00

Assume you have two independent, discrete random variables X and Y, which follow independent symmetric Bernoulli distributions taking only the values $+1$ and -1, both with probability $1/2$. As in the lecture, let $\kappa_n(X)$ denote the n-th order cumulant of a random variable X. Compute the quantities below.

Hint: Recall that $\kappa_3(X) = \mathrm{E}(X^3)$ and $\kappa_4(X) = \mathrm{E}(X^4) - 3(\mathrm{E}(X^2))^2$ if X is centered.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $\kappa_3(X) =$

  ✖
- $\kappa_4(X) =$

  ✖
- $\kappa_4(3X)/\kappa_4(X) =$

  ✖
- $\kappa_4(X+Y)/\kappa_4(X) =$

  ✖

Question **6**

Not answered

Marked out of 6.00

Which statements about Principal Component Analysis (PCA) are correct?

Multiple answers are possible!

Select one or more:

- a. PCA is a data projection method that aims at minimizing the variance in the projected data on a lower dimensional space.
- b. Common applications of PCA include data compression, data visualization and data pre-processing.
- c. The matrix of eigenvectors $\mathbf{U}$ is symmetric and positive-semidefinite.
- d. The eigenvector corresponding to the smallest eigenvalue of the eigendecomposition of the sample covariance matrix is called the first principal component.
- e. The sample mean of the k-th projection is equal to the k-th eigenvalue of the sample covariance matrix.
- f. One can use the singular value decomposition of the data matrix $\mathbf{X}$ to perform PCA.
- g. The projected data is given by $\mathbf{Y} = \mathbf{XU}$, where $\mathbf{X}$ is the (centered) original data and $\mathbf{U}$ is the matrix of eigenvectors.
- h. The PCA solution is unique up to signs, given that eigenvalues correspond to unique eigenvectors.

Die Antwort ist falsch.

The correct answers are: Common applications of PCA include data compression, data visualization and data pre-processing. ,
The projected data is given by $\mathbf{Y} = \mathbf{XU}$, where $\mathbf{X}$ is the (centered) original data and $\mathbf{U}$ is the matrix of eigenvectors.
, One can use the singular value decomposition of the data matrix $\mathbf{X}$ to perform PCA.
, The PCA solution is unique up to signs, given that eigenvalues correspond to unique eigenvectors.

Question **7**

Not answered

Marked out of 6.00

PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$\mathbf{X} = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix}$$

The eigenvectors of the covariance matrix of this data are:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(-1, 0, 1)^T$$
$$\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^T$$
$$\mathbf{u}_3 = (0, 1, 0)^T$$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order.

Compute the PCA projection (matrix) $\mathbf{Y}$.

Enter the components of the projected third sample $\mathbf{y}_3$ into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $y_3^{(1)} =$
  ✖
- $y_3^{(2)} =$
  ✖
- $y_3^{(3)} =$
  ✖

Question **8**

Not answered

Marked out of 6.00

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by the table below:

	$Y=0$	$Y=1$
$X=0$	0.25	0.125
$X=1$	0.25	0.375

This means, that e.g. $P(X=0, Y=0) = 0.25$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below.
Use the **natural logarithm** in your calculations.

You can use a pocket calculator. Round the final values to two decimal places. (You may want to keep three-digit accuracy if you use intermediate results.)
Number format example: 0.67

- $P(X=1) =$ ✖
- $P(Y=1) =$ ✖
- $H(X) =$ ✖
- $H(Y) =$ ✖
- $I(X;Y) =$ ✖

◄ Exam 1, Part A1

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Exam 1, Part B ►